APSC 261

Lecture 1

Technology: "The sum of all human making and devising and its application to society"

Society: "The totality of people who share territory, are organized through patterns, and make up a complex whole of shared meanings and technology"

Society is characterized by:

1. How people are organized
2. How people interact
3. What people know and believe

Domains society interfaces with:

1. Individual
2. Cultural
3. Social

Factors that relate to human thought / action / behaviour:

1. Self
2. Biological
3. Psychological
4. Behavioural
5. Situational

Functions of society:

1. Provides the needs for satisfaction
2. Helps individuals learn who they are
3. Helps individuals learn who others are and the characteristics of society

1 | Neutrality of Technology [Balabanian]

Premises that underly society:

1. Self-seeking
2. Elastic wants
3. Dominating nature
4. Neutrality of technology
5. Freedom of choice

Neutrality of Technology:

(False Premise) → "Tech is morally and politically neutral. Man is to blame for harm"

Freedom of choice:

(False premise) → "Individuals in the free market have total autonomy over purchases"

2 | True Grand Challenge: Self-knowledge [Mitcham]

Main Argument:

"We are entering a new axial age and the true challenge for engineers is self-reflection and exposure to the arts and humanities"

Lecture 2

Social technical divide:

"Technical realm recognized as legitimate source whereas Social Sciences are not

Barriers of the divide:

1. Engineering myths
2. Traditional perspective
3. Priviledging certain ideas over others

Characteristics of ENG Culture:

1. Prioritize economic over social
2. Isolated from other disciplines
3. Male dominated

Lecture 3

Why Context matters:

1. How we make sense of the world
2. How we act in the world

Freedom of choice:

"Illusion → technology mediates action"

Psychological Mechanisms:

1. Schema
2. Dual-Process theory

Schema:

"Represents a way to simplify thought process"

Dual-Process Theory:

"Human decision making relies on Deliberative and automatic mechanisms

Totality of factors (Complexity):

1. Unintended users
2. Intended uses
3. Technology breeds technology

3 | Technology choice and the good life [Dotson]

Technological Liberalism:

"Fallacy → technology extends human choice and benefit at no cost"

Technology Vs focal things:

Tech → Smartphone, TV

Focal → Fire(hearth), community square

Lecture 4/5

Dimensions of society:

1. Economic
2. values
3. Cultural
4. Technological
5. Social
6. Political

Culture:

"The complex whole of shared understandings and meanings by members of society"

Key features of culture:

1. Patterns
2. Repetition
3. Cumulative Effect
4. Subcultures
5. Webs of meaning

Cultural norms:

1. Folkways
2. Mores
3. Taboos
4. Laws

Beliefs:

"Assumptions about how things will be"

Values:

"How things ought to be"

4 | Social Responsibility of engineers [McFarland]

Obligation to come to aid:

1. Proximity
2. Ability to help
3. Lack of other help
4. Critical need

Technical Paternalism:

"Technical problems require technical solutions"

Lecture 6

Technological determinism:

"Technologies are neutral objects and culture plays no role

Social shaping:

"Too much focus on the user, overlooks structural factors"

Material vs non-material:

"Material is what you can see (physical objects). Non material refers to social processes, symbols and frameworks

Manifest vs Latent function:

Manifest → intended, expected beneficial

Latent → Unconscious, unintended and harmful

Framing technology as neutral:

"Car example, not that simple there are Latent unseen consequences

6 | Normative challenges of soft impacts [Swiersta]

Soft vs hard impacts:

"Hard impacts are tangible impacts that can be rectified. Soft impacts are more subtle but nonetheless important"

Norms and Values:

"Exist in practices (know-how) and are influenced by technology"

Lecture 7/8

Sustainability:

"Meets the needs of the present without compromising the needs of the future"

Needs theory:

"Meeting human needs is required for optimal development and functioning"

Socio-Cultural Environments:

1. Supportive
2. Need thwarting
3. Low support/depriving

Human needs and dependency:

1. Family
2. Social groups and peers
3. Societal Institutions

Need Satisfaction:

1. Physical/Survival
2. Social
3. Psychological

Vulnerability:

"Technology makes us vulnerable"

7 Centrality of human needs [Gough]

Needs vs Wants:

"Wants may be impulsive pleasures whereas needs are essential"

Universal needs theory:

"Basic needs are timeless and passed down by each generation

8 Empathy machines: Forgetting the body [Turkle]

Empathy machines:

"Unworthy machines that haven't lived a human life and yet we interact with them as if they could share our experience

Assault on empathy:

"The silent spring meaning people are glued to their phones and are not engaging in meaningful dialogue

Frictionless life:

"Technology has made it too easy to avoid hardship"

Lecture 9

Social stratification:

"How individuals are ranked and ordered in society"

Agents of social change:

1. Technology
2. Social institutions
3. Population
4. Environment

Types of Societies:

1. Hunter/Gatherer
2. Pastoral
3. Agricultural
4. Feudal
5. Industrial

Digital divide:

1. Unequal access
2. Unequal skills
3. Unequal outcomes

Sources of data collection:

1. Blogs
2. Social media
3. Sensors
4. Transactional data

Problems of big data:

1. Decisions are made for us
2. Biased algorithms
3. Discriminatory algorithms
4. Secretive and unaccountable
5. Our identities are for sale

9 Designing a good life: Matrix for technological mediation [Waelbers/Swierstra]

Matrix Axis:

x → Is, Can, Ought

y → Stakeholders, Consequences, good life

Matrix Categories:

1. Prescence
2. Empowerment
3. Rights
4. Anticipatory knowledge
5. Practical affordances
6. Responsibilities
7. Contingency
8. Freedom
9. Flourishing

Lecture 10

Technology Creep:

"Over time, technology morphs into something that was never intended at the time of its inception"

Lecture 11

What Engineers can do:

1. Due dilligence
2. Design practices
3. Join conversation

Definitions

Autonomy:
"Freedom from external control or influence"

Agency:
"The capacity to act or exert power"

Dialectic:
"The existence or action of opposing social forces"

Context:
"The circumstances that form the setting where an idea can be fully understood"

Direct:
"Aim something in a particular direction"

Disembody:
"Seeming to not have a body"

Dimension:
"A part or feature or way of considering something"

Domain:
"An area of interest or control"

Embed:
"Make something an integral part of"

Embody:
"To have/show particular qualities or ideas"

Hermeneutics:
"A method or theory of interpretation"

Heterogeneity:
"Quality or state of being diversion character or content"

Heuristic:
"Enabling a person to learn something for themselves"

Ideological:
"Based or relating to a system of ideas"

Interaction:
"Mutual/reciprocal action or influence"

Mediate:
"Occupy a middle position"

Neutral:
"No strong characteristics or features

Nuance:
"A small difference in feeling"

Ontology:
"set of concepts in a subject area"

Praxis:
"Accepted practice or custom"

Priviledge:
"A special right granted to a particular group"

Organize:
"Arrange something according to a system"

Organization:
"Form a coherent functioning whole"

Produce:
"Cause to happen"

Reify:
"Give definite form to a concept or idea"

Structure:
"Construct or arrange according to a plan"

Tacit:
"Understood without being stated"

Unencumbered:
"Not having burden or impediment"

Untenable:
"Not able to be defended"

Volition:
"The power of choosing or determining will"